to diminishing returns. Most mid-range models support 690,000 pixels effective or higher. However, an effective pixel count of 340,000 should suffice for most situations.

1.2.2. Illumination Rating And Low light Shooting

Most camcorders have trouble shooting in low light. Low-light video will typically have a grainy appearance, which is technically known as noise. Technically, anything other than bright sunlight is a low-light environment! Normal room lighting which is perfectly acceptable to the human eye may show up grainy on recorded video. Be sure to test out the low-light performance capabilities with the demo piece in the store. Shooting in environments such as restaurants will give you unwatchable video if the camcorder's low-light performance is not up to speed.

The illumination rating is the amount of ambient light required by the camcorder to operate without an additional light source. This is measured in lux, a measurement standard for lighting. Lower lux values are better. Entry-level models start at 7 lux or lower. Mid-range and higher models go down as far as 1 or 2 lux. However, this number can be deceptive, as the manufacturer may boast a lower lux level without mentioning anything about the quality of the video being recorded at that lux level! Additionally, if you're using a high-speed shutter, you require still higher illumination.

Another factor that affects the quality of low-light shoots is the aperture rating. Camcorders have iris diaphragms (apertures) that automatically adjust and control the amount of light reaching the CCD. This aperture rating, or F-stop, is denoted by numbers like f2.0, f1.4, or f1.3. The lower the rating, the better the camcorder will work in low light (see §1.2.6 (Shutter Speed) and §1.2.7 (Aperture) for more details).